Math 2C03 - Assignment 3

Fall 2026

This is one generated instance of the assignment. Your Childsmath questions may have different numerical values or choice orders. Submit your answers in Childsmath.

Question 1

Consider the following statements.

(i) Every linear nth order homogeneous ODE has a solution of the form $y(x) = e^{rx}$, where r is some real number.

(ii) Consider $y''' + ay'' + by' + cy = 0$ and let r_1, r_2, r_3 be the distinct roots (i.e., all are different) of the polynomial $P(r) = r^3 + ar^2 + br + c$. Then the general solution is given by $y(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x} + c_3 e^{r_3 x}$.

(iii) The fundamental set of solutions to $y'' - 2my' + m^2 y = 0$, where $m > 0$, is given by $\{e^{r_1 x}, e^{r_2 x}\}$ for some appropriate real values $r_1, r_2 > 0$.

(iv) The auxiliary equation of $ny'' - my' + ky = 0$, where n, m, k are real numbers with $n \neq 0$, is given by $nr^2 - mr + kr = 0$.

(v) Two complex values that are not real solve the auxiliary equation of $y'' + my' + m^2 y = 0$, where $m > 0$.

Determine which of the above statements are True (1) or False (2).

So, for example, if you think that the answers, in the above order, are True, False, False, True, False, then you would enter '1,2,2,1,2' into the answer box below (without the quotes).

Question 2

Which of the following sets of functions are linearly independent on the interval $(-\infty, \infty)$?

(i) $f_1(x) = x$, $f_2(x) = 3x$, $f_3(x) = x^2 + 7$.

(ii) $f_1(x) = 3e^{4x}$, $f_2(x) = 9e^{-4x}$, $f_3(x) = e^{4x} - e^{-4x}$.

(iii) $f_1(x) = 9\sin^2 x$, $f_2(x) = 4\cos^2 x$, $f_3(x) = 7$.

(A) (ii) only	(B) (ii) and (iii) only
(C) none of them	(D) (iii) only
(E) all of them	(F) (i) and (ii) only
(G) (i) and (iii) only	(H) (i) only

Question 3

For which of the cases below do the given functions form a fundamental set of solutions of the corresponding differential equation on the indicated interval?

(i) $y'' - 10y' + 21y = 0$; $8e^{5x}$, $5e^{3x}$, $(-\infty, \infty)$.

(ii) $y'' - 10y' + 25y = 0$; $2e^{5x}$, $6xe^{5x}$, $(-\infty, \infty)$.

(iii) $x^2y'' - 8xy' + 20y = 0$; $9x^4$, $3x^5$, $(0, \infty)$.

(A) (i) and (iii) only	(B) (i) and (ii) only
(C) (iii) only	(D) (ii) and (iii) only
(E) none of them	(F) all of them
(G) (i) only	(H) (ii) only

Question 4

Let

$$L(y) = a_n(x)y^{(n)}(x) + a_{n-1}(x)y^{(n-1)}(x) + \cdots + a_1(x)y'(x) + a_0(x)y(x),$$

where $a_0(x), a_1(x), \ldots, a_n(x)$ are known functions. Consider the nth order linear differential equation

$$L(y) = 9e^{6x}\cos x + 4xe^{6x}. \qquad (*)$$

Suppose that it is known that

$$\begin{aligned} L[y_1(x)] &= 6xe^{6x} && \text{when } y_1(x) = 30xe^{6x}, \\ L[y_2(x)] &= 12e^{6x}\sin x && \text{when } y_2(x) = 60e^{6x}\cos x, \\ L[y_3(x)] &= 11e^{6x}\cos x && \text{when } y_3(x) = 77e^{6x}\cos x + 231e^{6x}\sin x. \end{aligned}$$

Find a particular solution to $(*)$.

Question 5

Solve the following initial value problem.

$$y'' - 14y' + 149y = 0, \qquad y(0) = 9, \quad y'(0) = 10.$$

Question 6

Solve the following initial value problem.

$$y''' - 14y'' + 32y' + 128y = 0, \qquad y(0) = 0, \quad y'(0) = 0, \quad y''(0) = 4.$$

Question 7

Use Euler's method with step size $h = 0.1$ to approximate $y(4.2)$ where $y(x)$ is the solution to the following initial value problem.

$$y' = 2x + 3y + 6, \qquad y(4) = 9.$$

Question 8

(a) Find the largest interval on which one of the theorems given in class guarantees that the following initial value problem has a unique solution.

$$(x+9)y''' + (x^2 - 36)y'' + 19y = \frac{1}{x-4},$$
$$y(0) = 4, \qquad y'(0) = 9, \qquad y''(0) = 10.$$

(A) $(9, \infty)$	(B) $(4, \infty)$	(C) $(-9, \infty)$
(D) $(-4, \infty)$	(E) $(4, 9)$	(F) $(-\infty, -9)$
(G) $(-9, 4)$	(H) $(-\infty, 4)$	(I) $(-4, 9)$
(J) $(-\infty, 9)$	(K) $(-\infty, -4)$	(L) $(-9, -4)$

(b) Find the largest interval on which one of the theorems given in class guarantees that the following initial value problem has a unique solution.

$$(x-9)y''' + (x^2 - 36)y'' + 19y = \frac{1}{x-4},$$
$$y(0) = 4, \qquad y'(0) = 9, \qquad y''(0) = 10.$$

(A) $(-9, 4)$	(B) $(9, \infty)$	(C) $(-\infty, 9)$
(D) $(-9, \infty)$	(E) $(-4, 9)$	(F) $(-9, -4)$
(G) $(4, 9)$	(H) $(4, \infty)$	(I) $(-\infty, 4)$
(J) $(-\infty, -4)$	(K) $(-\infty, -9)$	(L) $(-4, \infty)$